



# João Roque Costa

Coimbra, Portugal

A Master's graduate with a strong interest in **problem-solving**, **financial markets**, **artificial intelligence**, and **data science**; eager to apply my theoretical knowledge in practical settings. Looking to join a dynamic team, develop new skills, and tackle real challenges in the IT industry.

## WORK EXPERIENCE

**Invited Teaching Assistant**  
**Instituto Superior Técnico**

Oct 24 - Jul 25

Taught and evaluated lab shifts on **Databases** and **Data Science**, covering SQL, database design and management using PostgreSQL and pgAdmin on the prior, and data treatment, visualization and interpretation using Python libraries such as **Pandas** and **Matplotlib** on the posterior.

## EDUCATION

**M. Sc. Computer Science and Engineering**  
**Instituto Superior Técnico**

Sep 23 - Oct 25

Focused in text **Artificial Intelligence** and **Data Science**, developed advanced skills in **Machine Learning**, **Deep Learning**, **Autonomous Agent and Multi-Agent Systems**, **Data Visualization**, **Data Science**, and **Cloud Computing**, but also **Optimization and Algorithms**.

To finalize the studies, I undertook a research-oriented thesis on **Maximum Variance Unfolding on Disjoint Manifolds**, which enlarged the method's **dimensionality reduction** capability into previously infeasible domains.

Additionally, a publication for **International Conference on Learning Representations (ICLR) 2026** was submitted.

Some technologies and tools I worked with include: **AWS**, **Python** (Sklern, Pandas, Numpy, Matplotlib, TensorFlow, Keras), **SQL** (PostgreSQL, pgAdmin, Server Management), **CmapTools**, **Pentaho**, **ArchiMate**, **Camunda Modeler**, and **Enterprise Architecture** (Sparx).

**B. Sc. in Computer Science and Engineering**  
**Instituto Superior Técnico**

Sep 20 - Jul 23

Worked on various projects that strengthened both my technical expertise and teamwork skills. Key areas of study and practical experience include: **Mathematics and Foundations** (Calculus, Physics, Algebra, Discrete Mathematics, Statistics), **Low-Level Computing** (Distributed Systems, Computer Networks, Operating Systems), **Data Bases**, **Machine Learning**, and **Management**. The principal technologies and tools I utilized during my studies were: **PostgreSQL**, **pgAdmin**, **Python** (NumPy, Pandas), **C**, **C++**, and **Java**.

This experience taught me how to effectively distribute tasks, collaborate on complex projects, and develop persistence in problem-solving.

## CONTACTS

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🌐 www.linkedin.com/in/jarcosta  
📄 www.github.com/jarcosta

## SKILLS

Python

SQL

C

Java

C++

R

JavaScript

SQL Server Management Studio

PostgreSQL

pgAdmin

AWS

Pentaho

ArchiMate

Camunda Modeler

Enterprise Architect

## CERTIFICATIONS

### [Cambridge English](#) C1 Advanced Exam

Score: 171

### [ABC bootcamps](#) English summer course Edinburgh

A two-week intensive English course in Edinburgh, where I developed my independence, communication and cooperation skills.

### [Udemy](#) Game Development

2D game development online course ([certificate](#)).

## HOBBIES

### [Projects and Practical Experience](#)

In my free time, I have developed various personal projects, showcased on my **GitHub**, that involve **API** integrations, **data handling**, and **decision-making algorithms**. Additionally, I am self-taught in **Excel**, which I regularly use to streamline and automate tasks.

### [Sports](#)

Swimming, Cycling, Snowboarding, Basketball

## PRE-STUDIES EXPERIENCE

### Software Development Summer Intern [Maria Adelaide and Playsketch](#)

July 2018

Assisted the development of an **email automation system**, improving workflow efficiency.

After completing the work, a **game development course** was awarded, where I built a **Space Invaders** clone and a **Number Wizard** game for mobile devices.